

BFAI DocIntel Engine v2.0

Product Technical Specification Sheet

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1. Product Overview

The BFAI DocIntel Engine v2.0 is a cloud-native document intelligence microservice that provides OCR parsing, LLM-based classification, vector embedding, and agentic retrieval-augmented generation (RAG) over uploaded documents. It is designed for enterprise document workflows requiring high accuracy, citation traceability, and sub-2-second query latency at scale.

2. Technical Specifications

Parameter	Value
Supported Input Formats	PDF (text + scanned), PNG, JPG, JPEG, TIFF, TXT, MD, CSV
Max File Size	20 MB per file; 200 MB per batch request
Max Pages per Document	500 pages
OCR Engine	Tesseract 5.x with LSTM model; fallback to pdfplumber for text-layer PDFs
Classification Model	Groq llama-3.3-70b-versatile; structured JSON output with schema validation
Embedding Model	sentence-transformers/all-MiniLM-L6-v2; 384-dimensional vectors
Vector Store	ChromaDB v0.5.x; cosine similarity; persistent on-disk index
RAG Framework	LangChain v0.2.x; tool-calling agent; top-5 chunk retrieval
Chunk Size	800 tokens; 100-token overlap (RecursiveCharacterTextSplitter)
Query Latency (p50)	< 1.2 seconds (Groq API) + < 50ms ChromaDB retrieval
Query Latency (p99)	< 3.5 seconds under 50 concurrent users
TTS Engine	ElevenLabs eleven_multilingual_v2; streaming audio/mpeg
STT Engine	OpenAI Whisper-1; server-side fallback; 25 MB audio limit
Rate Limits	10 uploads/min; 30 chat queries/min; 20 TTS calls/min per IP
Authentication	API key header (X-API-Key) for production; open for demo deployment
Security	MIME validation, filename sanitisation, UUID-only doc IDs, security headers
Deployment	Docker on Google Cloud Run; asia-south1; auto-scales 0-10 instances
Memory Requirement	2 GiB RAM minimum (for sentence-transformers model loading)
API Protocol	REST/JSON over HTTPS; OpenAPI 3.1 spec at /docs

3. API Endpoints

Endpoint	Description
POST /api/upload	Upload 1-10 documents; returns doc_ids; triggers async processing
GET /api/processing-status/{doc_id}	Poll document processing state
GET /api/documents	List all indexed documents with metadata
GET /api/documents/{doc_id}	Get single document detail + classification
GET /api/documents/{doc_id}/page/{n}/image	Retrieve page PNG thumbnail
DELETE /api/documents/{doc_id}	Remove document from index and storage
POST /api/chat	Send message; returns RAG answer + citations
POST /api/voice/speak	TTS: text -> audio/mpeg stream
POST /api/voice/transcribe	STT: audio file -> transcript text

GET /api/health

Health check; returns indexed document count